

Discrete Structures

- Chapter 4, Number Theory
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- Spring 2026
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The divides operator

Definition

For $a, b \in \mathbb{Z}, a \neq 0$, we say $a \mid b : (a \text{ divides } b)$ if $\exists c \in \mathbb{Z} : b = ac$

■ $4 \mid 12$ ▷ $12 = 4 \cdot 3$

■ $3 \mid 12$ ▷ $12 = 3 \cdot 4$

■ $5 \mid 0$ ▷ $0 = 5 \cdot 0$

■ $3 \nmid 7$

■ $1 \mid 8$ ▷ $8 = 1 \cdot 8$

■ $-2 \mid 6$ ▷ $6 = -2 \cdot -3$

■ $-6 \mid -12$ ▷ $-12 = -6 \cdot 2$

■ $-4 \nmid 13$

■ a is a **factor** or **divisor** of b

■ b is a **multiple** of a

Divisibility Facts

$$1 \quad \forall n \quad 1 \mid n$$

$$\triangleright n = 1 \cdot n$$

$$2 \quad \forall n \quad n \mid n$$

$$\triangleright n = n \cdot 1$$

$$3 \quad \forall n \quad n \mid 0$$

$$\triangleright 0 = n \cdot 0$$

$$4 \quad \forall n \quad -1 \mid n$$

$$\triangleright n = -1 \cdot -n$$

$$5 \quad \forall n \quad -n \mid n$$

$$\triangleright n = -n \cdot -1$$

Divisibility Facts

Theorem

For $a, b, c \in \mathbb{Z}$

$$\textbf{1} \quad a \mid b \implies a \mid bc$$

$$\textbf{2} \quad a \mid b \wedge b \mid c \implies a \mid c$$

$$\textbf{3} \quad a \mid b \wedge a \mid c \implies a \mid b + c$$

$$\triangleright 3 \mid 6 \implies 3 \mid 6 \cdot 2$$

$$\triangleright 2 \mid 4 \wedge 4 \mid 8 \implies 2 \mid 8$$

$$\triangleright 2 \mid 4 \wedge 2 \mid 8 \implies 2 \mid 8 + 4$$

Corollary: $a \mid b \wedge a \mid c \implies a \mid mb + nc, \quad m, n \in \mathbb{Z}$

$$\triangleright 2 \mid 4 \wedge 2 \mid 8 \implies 2 \mid 3 \cdot 8 + 5 \cdot 4$$

Divisibility Facts

Corollary: $a \mid b \wedge a \mid c \implies a \mid mb + nc, \quad m, n \in \mathbb{Z}$

Proof: Number theory proofs generally use definition and basic arithmetic

$$a \mid b \wedge a \mid c \implies \exists x, y : \quad b = ax \quad \wedge \quad c = ay$$

$$mb = m(ax) = a(mx) \implies a \mid mb$$

$$nc = n(ay) = a(ny) \implies a \mid nc$$

By Theorem part (2) $a \mid mb + nc$

Congruence

For $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, $a \equiv_m b$ iff $m \mid (a - b)$

pronounced as a is congruent to b modulo m

▷ Standard notation for $a \equiv_m b$ is $a \equiv b \pmod{m}$

Theorem: Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$.

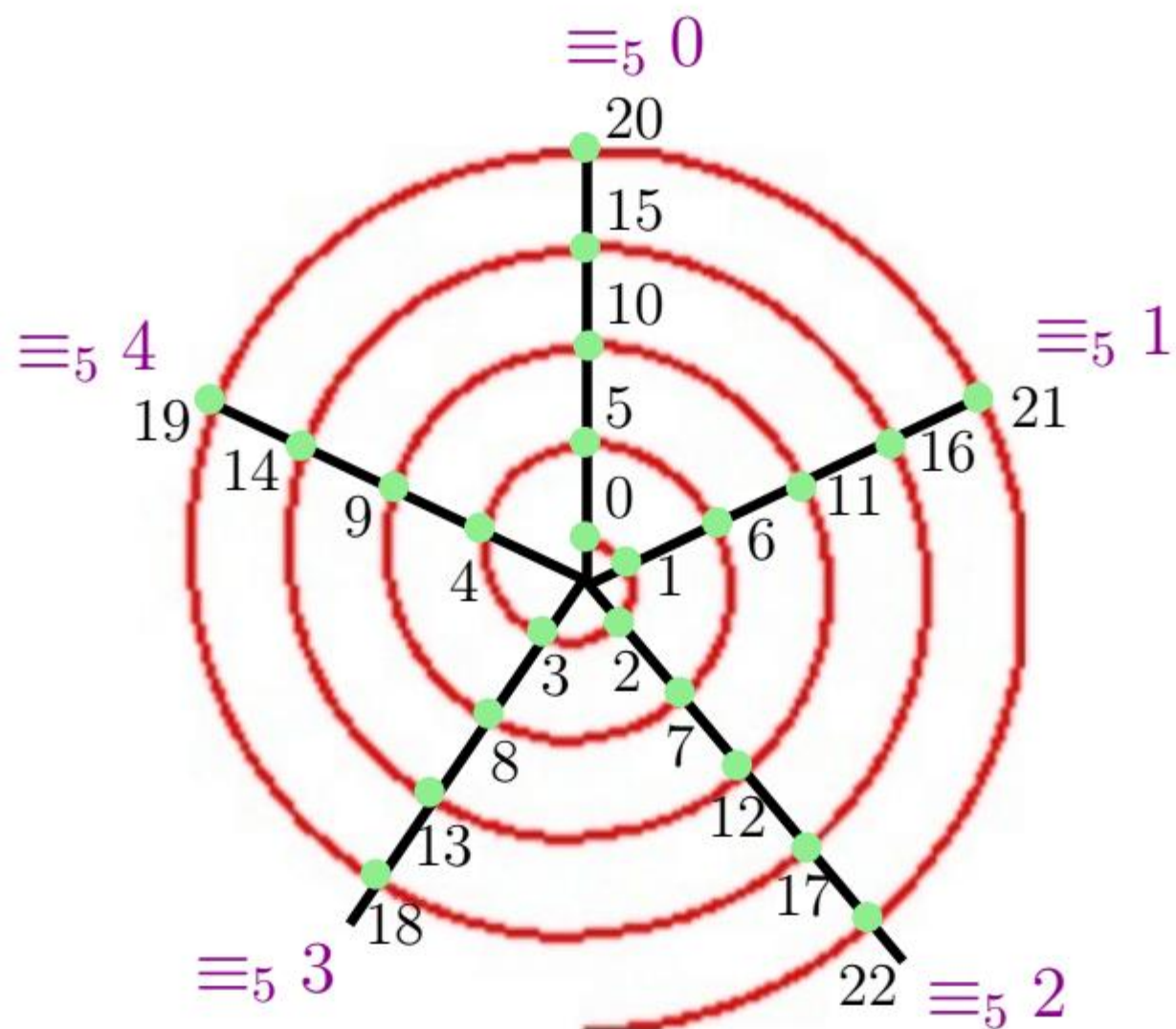
Then $a \equiv_m b$ iff $a \% m = b \% m$

$$3 \equiv_3 6, \quad 3 \equiv_3 3, \quad 7 \equiv_5 2 \quad -3 \equiv_5 2, \quad -1 \equiv_3 -4$$

To avoid confusion between standard notations - $\pmod{m}$ vs **mod** m , we use our notation.

Note that $\% m$ is an operator, while $\equiv_m$ is an equivalence relation over $\mathbb{Z}$

Congruence



Congruence Facts

Fact

1 $a \equiv_m a$

2 $a \equiv_m b \iff b \equiv_m a$

3 $a \equiv_m b \wedge b \equiv_m c \implies a \equiv_m c$

▷ $\equiv_m$ is an equivalence relation on $\mathbb{Z}$

4 $a \equiv_m (a \% m)$

Congruence

Theorem

$$a \equiv_m b \iff \exists k \in \mathbb{Z} : a = b + km$$

$$\triangleright 8 \equiv_5 3 \text{ and } 8 = 3 + 5(1)$$

$$\triangleright 16 \equiv_5 1 \text{ and } 16 = 1 + 5(3)$$

Proof:

$$a \equiv_m b$$

$$\iff m \mid (a - b)$$

$$\iff \exists k \in \mathbb{Z} : a - b = km$$

$$\iff a = b + km$$

$\triangleright$ by definition

Congruence

Definition

For $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, $a \equiv_m b$ iff $m \mid (a - b)$

Theorem

For $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, $a \equiv_m b$ iff $a \% m = b \% m$

Theorem

For $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$, $a \equiv_m b \iff \exists k \in \mathbb{Z} : a = b + km$

Modular Arithmetic

Lemma

If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$

$$\triangleright 8 \equiv_5 3 \text{ and } 9 \equiv_5 4 \implies 8 + 9 \equiv_5 3 + 4$$

Familiar cases: $m = 2$ and $m = 10$

If (a, b) and (c, d) have the same parity, then $a + c$ and $b + d$ have the same parity

If (a, b) and (c, d) have the same last digit, then $a + c$ and $b + d$ have the same last digit

The lemma says it works for all m

Modular Arithmetic

Lemma

If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$

Proof:

$$a \equiv_m b \implies a = b + xm \quad \text{AND}$$

$$c \equiv_m d \implies c = d + ym$$

$$a + c = b + d + xm + ym \implies (a + c) - (b + d) = m(x + y)$$

$$\text{Hence } m \mid (a + c) - (b + d)$$

$$\text{So } a + c \equiv_m b + d$$

Modular Arithmetic

Lemma

If $a \equiv_m b$ and $c \equiv_m d$, then $ac \equiv_m bd$

Proof:

Very similar!

Lemma

If $a \equiv_m b$, then $a^k \equiv_m b^k$

Proof:

Very similar!

Modular Arithmetic

Lemma

- 1 If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$
- 2 If $a \equiv_m b$ and $c \equiv_m d$, then $ac \equiv_m bd$
- 3 If $a \equiv_m b$, then $a^k \equiv_m b^k$

Corollary

- 1 $(a + b) \% m = ((a \% m) + (b \% m)) \% m$
- 2 $ab \% m = ((a \% m)(b \% m)) \% m$
- 3 $a^k \% m = (a \% m)^k \% m$

This means that while computing $(a + c) \% m$ or $(ac) \% m$, we can replace a with $(a \% m)$ and c with $(c \% m)$ $\triangleright$ Recall that $a \equiv_m a \% m$

Modular Arithmetic

Lemma

- 1 If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$
- 2 If $a \equiv_m b$ and $c \equiv_m d$, then $ac \equiv_m bd$
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Compute $-706 \cdot 1456 \% 19$

$$-706 \equiv_{19} 16 \text{ and } 1456 \equiv_{19} 12 \implies -706 \cdot 1456 \% 19 = 16 \cdot 12 \% 19$$

Modular Arithmetic

Lemma

- 1 If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$
- 2 If $a \equiv_m b$ and $c \equiv_m d$, then $ac \equiv_m bd$
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Corollary

- 1 $(a + b) \% m = ((a \% m) + (b \% m)) \% m$
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$A = \{-706, 1456, 88, -41, 19, 20, 38, 40\}$ Compute $\left(\sum_{x \in A} x\right) \% 19$

Remainders: $R = \{16, 12, 12, 16, 0, 1, 0, 2\}$ So $\left(\sum_{x \in A} x\right) \% 19 = \left(\sum_{r \in R} r\right) \% 19$

Modular Arithmetic

Lemma

- 1 If $a \equiv_m b$ and $c \equiv_m d$, then $a + c \equiv_m b + d$
- 2 If $a \equiv_m b$ and $c \equiv_m d$, then $ac \equiv_m bd$
- 3 If $a \equiv_m b$, then $a^k \equiv_m b^k$

Corollary

- 1 $(a + b) \% m = ((a \% m) + (b \% m)) \% m$
- 2 $ab \% m = ((a \% m)(b \% m)) \% m$
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Compute $516^{3031} \% 103$

$$516 \equiv_{103} 1 \quad \text{So } 516^{3031} \% 103 = 1^{3031} \% 103 = 1$$

Modular Exponentiation

Find $22^4 \% 29$

Notice that we can take $\%$ after each multiplication

$$22^4 \% 29 = 22 \cdot 22 \cdot 22 \cdot 22 \% 29$$

$$= 22 \cdot 22 \cdot 484 \% 29 = 22 \cdot 22 \cdot 20 \% 29$$

$$= 22 \cdot 440 \% 29 = 22 \cdot 5 \% 29 = 110 \% 29 = 23$$

Prime Numbers

Definition

A positive integer p is prime if it has exactly two divisors, namely 1 and p

1 is not prime

Definition

A positive integer n is composite if it has a divisor d , $1 < d < n$

1 is not composite

Greatest common divisor

$\text{GCD}(a, b) :=$ the greatest common divisor

▷ the largest integer d that divides both a and b

$$\text{GCD}(24, 32) = 8$$

$$\text{GCD}(22, 24) = 2$$

$$\text{GCD}(15, 5) = 5$$

$$\text{GCD}(25, 15) = 5$$

$$\text{GCD}(13, 20) = 1$$

$$\text{GCD}(11, 33) = 11$$

Lemma: p is prime $\implies \forall a \in \mathbb{Z} \text{ GCD}(p, a) = 1 \text{ or } p$

▷ $\because p$ has only two divisors 1 and p

Greatest common divisor

$\text{GCD}(a, b) :=$ the greatest common divisor

▷ the largest integer d that divides both a and b

a and b are **relatively prime** if $\text{GCD}(a, b) = 1$

Equivalently, a and b have no common factors

$$\text{GCD}(25, 16) = 1, \quad \text{GCD}(24, 25) = 1$$

A prime number p is relatively prime to all integers except its multiples

Greatest common divisor

$\text{GCD}(a, b) :=$ the greatest common divisor

▷ the largest integer d that divides both a and b

We can find $\text{GCD}(a, b)$ by

finding all divisors of a and b , then

find the common divisors, and then

find the greatest among the commons

Greatest common divisor

$\text{GCD}(a, b) :=$ the greatest common divisor

▷ the largest integer d that divides both a and b

We can find $\text{GCD}(a, b)$ from the prime factorization of a and b

$$a = p_1^{a_1} p_2^{a_2} \dots p_n^{a_n} \qquad b = p_1^{b_1} p_2^{b_2} \dots p_n^{b_n}$$

$$\text{GCD}(a, b) = p_1^{\min\{a_1, b_1\}} p_2^{\min\{a_2, b_2\}} \dots p_n^{\min\{a_n, b_n\}}$$

$$98 = 2 \cdot 7 \cdot 7 = 2^1 3^0 5^0 7^2 11^0 \dots$$

$$420 = 2 \cdot 2 \cdot 3 \cdot 5 \cdot 7 = 2^2 3^1 5^1 7^1 11^0 \dots$$

$$\text{GCD}(98, 420) = 2^1 3^0 5^0 7^1 11^0 \dots = 14$$